

Lösungen Übungsblatt 0a (Repetition Terme)

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Ausmultiplizieren

1. Aufgabe

Vereinfache die folgenden Terme indem du die Klammern auflöst:

(a)

$$(3x + 2)(2x - 3)(x - 1) - (x - 1)(x + 2)(x - 3) = ?$$

Lösung:

$$\begin{aligned} \dots &= (x - 1) [(6x^2 - 5x - 6) - (x^2 - x - 6)] \\ &= (x - 1) (5x^2 - 4x) \\ &= 5x^3 - 9x^2 + 4x \end{aligned}$$

(b)

$$(3x + 2)(5x + 7) - \{-[-3(5x - 3)(7x + 5) - (2x + 8)(3x + 7)]\} = ?$$

Lösung:

$$\begin{aligned} \dots &= 15x^2 + 31x + 14 + [-105x^2 - 12x + 45 - 6x^2 - 38x - 56] \\ &= 15x^2 + 31x + 14 + -105x^2 - 12x + 45 - 6x^2 - 38x - 56 \\ &= -96x^2 - 19x + 3 \end{aligned}$$

(c)

$$-a - (-b) + \{-3a - [-7 + (2a - 3b)]\} = ?$$

Lösung:

$$\begin{aligned} -a - (-b) + \{-3a - [-7 + (2a - 3b)]\} &= -a - (-b) + \{-3a - [-7 + 2a - 3b]\} \\ &= -a - (-b) + \{-3a + 7 - 2a + 3b\} \\ &= -a - (-b) + \{7 - 5a + 3b\} \\ &= -a + b + 7 - 5a + 3b \\ &= 7 - 6a + 4b \end{aligned}$$

(d)

$$(3x - 7)^2 - (3x + 1)(3x - 1) = ?$$

Lösung:

$$\begin{aligned} (3x - 7)^2 - (3x + 1)(3x - 1) &= (9x^2 - 42x + 49) - (9x^2 - 1) \\ &= 9x^2 - 42x + 49 - 9x^2 + 1 \\ &= 50 - 42x \end{aligned}$$

(e)

$$(1 - a^3)^3 + (a^3 - 1)^3 = ?$$

Lösung:

$$\begin{aligned}(1 - a^3)^3 + (a^3 - 1)^3 &= (1 - 3a^3 + 3a^6 - a^9) + (a^9 - 3a^6 + 3a^3 - 1) \\ &= 0\end{aligned}$$

(f)

$$(1 - a)^2 - ((a - 1)^2 - (a^2 - 1)) = ?$$

Lösung:

$$\begin{aligned}(1 - a)^2 - ((a - 1)^2 - (a^2 - 1)) &= 1 - 2a + a^2 - ((a^2 - 2a + 1) - (a^2 - 1)) \\ &= 1 - 2a + a^2 - (a^2 - 2a + 1 - a^2 + 1) \\ &= 1 - 2a + a^2 - (2 - 2a) \\ &= 1 - 2a + a^2 - 2 + 2a \\ &= a^2 - 1\end{aligned}$$

(g)

$$a^2b^2 - (a^2 - 1)(b^2 + 1) - (a + b)^2 = ?$$

Lösung:

$$\begin{aligned}a^2b^2 - (a^2 - 1)(b^2 + 1) - (a + b)^2 &= a^2b^2 - (a^2b^2 + a^2 - b^2 - 1) - (a^2 + 2ab + b^2) \\ &= a^2b^2 - a^2b^2 - a^2 + b^2 + 1 - a^2 - 2ab - b^2 \\ &= 1 - 2ab - 2a^2\end{aligned}$$

(h)

$$-((2 - a)(3 - b) - ((4 - c)(5 - d) - (a + c)(b - d))) = ?$$

Lösung:

$$\begin{aligned}\dots &= -((6 - 3a - 2b + ab) - ((20 - 5c - 4d + cd) - (ab - ad + cb - cd))) \\ &= -(6 - 3a - 2b + ab - (20 - 5c - 4d + cd - ab + ad - cb + cd)) \\ &= -(6 - 3a - 2b + ab - 20 + 5c + 4d - 2cd + ab - ad + cb) \\ &= 14 + 3a + 2b - 2ab - 5c - 4d + 2cd + ad - cb\end{aligned}$$

Faktorisieren

2. Aufgabe

Faktoriere soweit wie möglich:

(a)

$$1 - 2a + a^2 - b^2 = ?$$

Lösung:

$$1 - 2a + a^2 - b^2 = (1 - a)^2 - b^2 = ((1 - a) - b)((1 - a) + b) = (1 - a - b)(1 - a + b)$$

(b)

$$6a + 3ac - 2b - bc = ?$$

Lösung:

$$6a + 3ac - 2b - bc = 3a(2 + c) - b(2 + c) = (2 + c)(3a - b)$$

(c)

$$(x - 2)(2a + 3b) - (a - b)(x - 2) + (b + a)(2 - x) = ?$$

Lösung:

$$\begin{aligned}(x - 2)(2a + 3b) - (a - b)(x - 2) + (b + a)(2 - x) &= \\(x - 2)(2a + 3b) - (x - 2)(a - b) - (x - 2)(b + a) &= \\(x - 2)[(2a + 3b) - (a - b) - (b + a)] &= \\(x - 2)[2a + 3b - a + b - b - a] &= 3b(x - 2)\end{aligned}$$

(d)

$$x - y - x^2 + 2xy - y^2 = ?$$

Lösung:

$$\begin{aligned}x - y - x^2 + 2xy - y^2 &= \\x - y - (x^2 - 2xy + y^2) &= \\(x - y) - (x - y)^2 &= \\(x - y)[1 - (x - y)] &= (x - y)(1 - x + y)\end{aligned}$$

(e)

$$u^2 - 14u + 49 = ?$$

Lösung:

$$\begin{aligned}u^2 - 14u + 49 &= \\u^2 - 2 \cdot 7 \cdot u + 7^2 &= (u - 7)^2\end{aligned}$$

(f)

$$12 - 8z + z^2 = ?$$

Lösung:

$$\begin{aligned}12 - 8z + z^2 &= \\z^2 - 8z + 12 &= \\z^2 - 6z - 2z + 12 &= \\z(z - 6) - 2(z - 6) &= (z - 6)(z - 2)\end{aligned}$$

(g)

$$16 - x^2 + 4xz^2 - 4z^4 = ?$$

Lösung:

$$\begin{aligned}
16 - x^2 + 4xz^2 - 4z^4 &= \\
16 - (x^2 - 4xz^2 + 4z^4) &= \\
16 - (x^2 - 2 * x * (2z^2) + (2z^2)^2) &= \\
4^2 - (x - 2z^2)^2 &= \\
[4 - (x - 2z^2)] [4 + (x - 2z^2)] &= (4 - x + 2z^2) (4 + x - 2z^2)
\end{aligned}$$

(h)

$$(a - b)(2x + 3y) - (2x - y)(a - b) - (b - a)(x - 3y) = ?$$

Lösung:

$$\begin{aligned}
\dots &= (a - b)(2x + 3y) - (2x - y)(a - b) + (a - b)(x - 3y) \\
&= (a - b)((2x + 3y) - (2x - y) + (x - 3y)) \\
&= (a - b)(2x + 3y - 2x + y + x - 3y) \\
&= (a - b)(x + y)
\end{aligned}$$

(i)

$$xy - x - yz - y + z + 1 = ?$$

Lösung:

$$xy - x - yz - y + z + 1 = x(y - 1) - z(y - 1) - (y - 1) = (y - 1)(x - z - 1)$$

(j)

$$d^2 + 8d + 15 = ?$$

Lösung:

$$d^2 + 8d + 15 = (d + 3)(d + 5)$$

(k)

$$96a^3x + 48a^2x^2 + 6ax^3 = ?$$

Lösung:

$$96a^3x + 48a^2x^2 + 6ax^3 = 6ax(16a^2 + 8ax + x^2) = 6ax(4a + x)^2$$

(l)

$$x^3 - 4x = ?$$

Lösung:

$$x^3 - 4x = x(x^2 - 4) = x(x - 2)(x + 2)$$

(m)

$$4a^2 - b^2 + 2bc - c^2 = ?$$

Lösung:

$$4a^2 - b^2 + 2bc - c^2 = 4a^2 - (b - c)^2 = (2a - (b - c))(2a + (b - c)) = (2a - b + c)(2a + b - c)$$

(n)

$$3x^2y + 3xy - 60y = ?$$

Lösung:

$$3x^2y + 3xy - 60y = 3y(x^2 + x - 20) = 3y(x + 5)(x - 4)$$

(o)

$$81a^4 - 1 = ?$$

Lösung:

$$81a^4 - 1 = (9a^2 - 1)(9a^2 + 1) = (3a - 1)(3a + 1)(9a^2 + 1)$$

(p)

$$9 - 6x + x^2 = ?$$

Lösung:

$$9 - 6x + x^2 = (3 - x)^2$$

(q)

$$a^2 - b^2 + 2b - 1 = ?$$

Lösung:

$$\begin{aligned} a^2 - b^2 + 2b - 1 &= a^2 - (b^2 - 2b + 1) \\ &= a^2 - (b - 1)^2 \\ &= (a - (b - 1))(a + (b - 1)) \\ &= (a - b + 1)(a + b - 1) \end{aligned}$$

(r)

$$x^4 - 625 = ?$$

Lösung:

$$x^4 - 625 = (x^2 - 25)(x^2 + 25) = (x - 5)(x + 5)(x^2 + 25)$$

(s)

$$a^2 - 3ax - 3bx + ab = ?$$

Lösung:

$$\begin{aligned} a^2 - 3ax - 3bx + ab &= a(a - 3x) + b(a - 3x) \\ &= (a - 3x)(a + b) \end{aligned}$$

Brüche

3. Aufgabe

Kürze die folgenden Brüche soweit wie möglich:

(a)

$$b_1 = \frac{13005}{5083} = ?$$

Lösung:

$$b_1 = \frac{13005}{5083} = \frac{3 * 3 * 5 * 17 * 17}{13 * 17 * 23} = \frac{3 * 3 * 5 * 17}{13 * 23} = \frac{765}{299}$$

(b)

$$b_2 = \frac{15a - 6ab}{20c - 8bc} = ?$$

Lösung:

$$b_2 = \frac{15a - 6ab}{20c - 8bc} = \frac{3a(5 - 2b)}{4c(5 - 2b)} = \frac{3a}{4c}$$

(c)

$$b_3 = \frac{a - x}{x - a} = ?$$

Lösung:

$$b_3 = \frac{a - x}{x - a} = \frac{-(x - a)}{x - a} = -1$$

(d)

$$b_4 = \frac{(4a - 1)^2}{1 - 16a^2} = ?$$

Lösung:

$$b_4 = \frac{(4a - 1)^2}{1 - 16a^2} = \frac{(1 - 4a)^2}{(1 - 4a)(1 + 4a)} = \frac{1 - 4a}{1 + 4a}$$

(e)

$$b_5 = \frac{x^2 - 9x - 10}{x^2 - 1} = ?$$

Lösung:

$$b_5 = \frac{x^2 - 9x - 10}{x^2 - 1} = \frac{(x + 1)(x - 10)}{(x + 1)(x - 1)} = \frac{x - 10}{x - 1}$$

(f)

$$b_6 = \frac{4x - x^3}{2x + x^2} = ?$$

Lösung:

$$b_6 = \frac{4x - x^3}{2x + x^2} = -\frac{x(x - 2)(x + 2)}{x(x + 2)} = -(x - 2) = 2 - x$$

4. Aufgabe

Berechne:

(a)

$$\frac{1}{x^2 - 2} + \frac{1}{2 - x} + \frac{1}{2 + x} = ?$$

Lösung:

$$\begin{aligned} \frac{1}{x^2 - 4} + \frac{1}{2 - x} + \frac{1}{2 + x} &= \frac{1}{(x - 2)(x + 2)} - \frac{1}{x - 2} + \frac{1}{x + 2} \\ &= \frac{1}{(x - 2)(x + 2)} - \frac{x + 2}{(x - 2)(x + 2)} + \frac{x - 2}{(x - 2)(x + 2)} \\ &= \frac{1 - (x + 2) + (x - 2)}{(x - 2)(x + 2)} \\ &= \frac{1 - x - 2 + x - 2}{(x - 2)(x + 2)} \\ &= -\frac{3}{(x - 2)(x + 2)} \end{aligned}$$

(b)

$$\frac{a}{a^2 - 5a + 4} - \frac{a + 4}{a^2 + 3a - 4} = ?$$

Lösung:

$$\begin{aligned} \frac{a}{a^2 - 5a + 4} - \frac{a + 4}{a^2 + 3a - 4} &= \frac{a}{(a - 1)(a - 4)} - \frac{a + 4}{(a - 1)(a + 4)} \\ &= \frac{a}{(a - 1)(a - 4)} - \frac{1}{(a - 1)} \\ &= \frac{a}{(a - 1)(a - 4)} - \frac{a - 4}{(a - 1)(a - 4)} \\ &= \frac{a - (a - 4)}{(a - 1)(a - 4)} \\ &= \frac{a - a + 4}{(a - 1)(a - 4)} \\ &= \frac{4}{(a - 1)(a - 4)} \end{aligned}$$

(c)

$$x + \frac{x}{x - a} + a + \frac{a}{a - x} = ?$$

Lösung:

$$\begin{aligned}
x + \frac{x}{x-a} + a + \frac{a}{a-x} &= x + \frac{x}{x-a} + a - \frac{a}{x-a} \\
&= \frac{x(x-a)}{x-a} + \frac{x}{x-a} + \frac{a(x-a)}{x-a} - \frac{a}{x-a} \\
&= \frac{x^2 - ax + x + ax - a^2 - a}{x-a} \\
&= \frac{x^2 - a^2 + x - a}{x-a} \\
&= \frac{(x-a)(x+a) + (x-a)}{x-a} \\
&= \frac{(x-a)(x+a+1)}{x-a} \\
&= x + a + 1
\end{aligned}$$

(d)

$$\frac{\frac{1}{x-\frac{1}{x}}}{\frac{x}{1+\frac{x}{1}}}=?$$

Lösung:

$$\frac{\frac{1}{x-\frac{1}{x}}}{\frac{x}{1+\frac{x}{1}}} = \frac{\frac{1}{\frac{x^2-1}{x}}}{\frac{x}{1+x}} = \frac{\frac{1}{x^2-1} \cdot x}{\frac{x}{1+x}} = \frac{\frac{x}{x^2-1}}{\frac{x}{1+x}} = \frac{x}{(x-1)(x+1)} \cdot \frac{x+1}{x} = \frac{1}{x-1}$$

(e)

$$\frac{ax+y}{a+n} + \frac{nx-y}{a+n}=?$$

Lösung:

$$\frac{ax+y}{a+n} + \frac{nx-y}{a+n} = \frac{ax+y+nx-y}{a+n} = \frac{x(a+n)}{a+n} = x$$

(f)

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}=?$$

Lösung:

$$\dots = \frac{60+30+20+15+12+10}{60} = \frac{147}{60} = \frac{49}{20}$$

(g)

$$\frac{5x+3y}{3a} - \frac{2x+5y}{6b} - \frac{8x+6y}{6ab}=?$$

Lösung:

$$\begin{aligned}
\frac{5x+3y}{3a} - \frac{2x+5y}{6b} - \frac{8x+6y}{6ab} &= \frac{2b(5x+3y) - a(2x+5y) - (8x+6y)}{6ab} \\
&= \frac{10bx + 6by - 2ax - 5ay - 8x - 6y}{6ab}
\end{aligned}$$

(h)

$$\frac{2x+5}{4x-4} + \frac{5x-3}{6x-6} + \frac{2x+3}{2-2x} = ?$$

Lösung:

$$\begin{aligned} \frac{2x+5}{4x-4} + \frac{5x-3}{6x-6} + \frac{2x+3}{2-2x} &= \frac{2x+5}{4(x-1)} + \frac{5x-3}{6(x-1)} - \frac{2x+3}{2(x-1)} \\ &= \frac{3(2x+5) + 2(5x-3) - 6(2x+3)}{12(x-1)} \\ &= \frac{6x+15+10x-6-12x-18}{12(x-1)} \\ &= \frac{4x-9}{12(x-1)} \end{aligned}$$

(i)

$$\frac{2n-3x}{3a+3} - \frac{5x+2n}{5b-5} + \frac{ax+bx}{ab-a+b-1} = ?$$

Lösung:

$$\begin{aligned} \dots &= \frac{2n-3x}{3(a+1)} - \frac{5x+2n}{5(b-1)} + \frac{ax+bx}{(a+1)(b-1)} \\ &= \frac{5(b-1)(2n-3x) - 3(a+1)(5x+2n) + 15(ax+bx)}{15(a+1)(b-1)} \\ &= \frac{10bn - 15bx - 10n + 15x - 15ax - 6an - 15x - 6n + 15ax + 15bx}{15(a+1)(b-1)} \\ &= \frac{2n(5b-3a-8)}{15(a+1)(b-1)} \end{aligned}$$

(j)

$$\frac{4a^2}{2a+b} + \frac{8ab^3}{(2a-b)(4a^2-b^2)} - \frac{b^2(2a+b)}{(2a-b)^2} = ?$$

Lösung:

$$\begin{aligned} \dots &= \frac{4a^2}{2a+b} + \frac{8ab^3}{(2a-b)^2(2a+b)} - \frac{b^2(2a+b)}{(2a-b)^2} = \frac{4a^2(2a-b)^2 + 8ab^3 - b^2(2a+b)^2}{(2a-b)^2(2a+b)} = \\ &= \frac{16a^4 - 16a^3b + 4a^2b^2 + 8ab^3 - 4a^2b^2 - 4ab^3 - b^4}{(2a-b)^2(2a+b)} = \frac{16a^4 - b^4 - 16a^3b + 4ab^3}{(2a-b)^2(2a+b)} = \\ &= \frac{(4a^2-b^2)(4a^2+b^2) - 4ab(4a^2-b^2)}{(2a-b)^2(2a+b)} = \frac{(4a^2-b^2)(4a^2-4ab+b^2)}{(2a-b)^2(2a+b)} = \\ &= \frac{(2a-b)^3(2a+b)}{(2a-b)^2(2a+b)} = 2a-b \end{aligned}$$

(k)

$$\frac{\frac{1}{a^2-1} - 1}{1 - \frac{1}{a+1}} = ?$$

Lösung:

$$\frac{\frac{1}{a^2-1} - 1}{1 - \frac{1}{a+1}} = \frac{\frac{2-a^2}{a^2-1}}{\frac{a}{a+1}} = \frac{(2-a^2)(a+1)}{a(a-1)(a+1)} = \frac{2-a^2}{a(a-1)}$$

(l)

$$\frac{\frac{x^2}{1-x^2} + 1}{\frac{x}{1-x} + 1} : \frac{1}{1+x} = ?$$

Lösung:

$$\frac{\frac{x^2}{1-x^2} + 1}{\frac{x}{1-x} + 1} : \frac{1}{1+x} = \frac{\frac{1}{1-x^2}}{\frac{1}{1-x}} \frac{1+x}{1} = \frac{(1-x)(1+x)}{(1-x)(1+x)} = 1$$

(m)

$$\frac{\frac{1}{x} + \frac{1}{y} + \frac{1}{z}}{\frac{1}{x} - \frac{1}{y} - \frac{1}{z}} : \frac{\frac{2x^2}{x^2-z^2}}{\frac{x}{x+y} + \frac{x}{x-z}} = ?$$

Lösung:

$$\begin{aligned} \dots &= \frac{\frac{yz+xz+xy}{xyz}}{\frac{yz-xz-xy}{xyz}} \frac{x(x-z) + x(x+y)}{(x+y)(x-z)} \frac{x^2-z^2}{2x^2} = \frac{yz+xz+xy}{yz-xz-xy} \frac{x(2x+y-z)}{(x+y)(x-z)} \frac{(x-z)(x+z)}{2x^2} = \\ &= \frac{x(yz+xz+xy)(2x+y-z)(x-z)(x+z)}{2x^2(yz-xz-xy)(x+y)(x-z)} = \frac{(yz+xz+xy)(2x+y-z)(x+z)}{2x(yz-xz-xy)(x+y)} = \end{aligned}$$